| | $\mathcal{K}^{\#1}_{0^+}$ | $\mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{K}^{\#1}_{0^+}\dagger$ | $-\frac{3\,\Upsilon_1\,k^2}{2}$ | 0 |
| $\mathcal{K}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$ | 0 | $3\,k^2\,\overset{(4)}{\kappa}_{15}$ |

| | $\mathcal{K}^{\#1}_{1^+\,\alpha\beta}$ | $\mathcal{K}^{\#2}_{1^+\,\alpha\beta}$ | $\mathcal{K}^{\#1}_{1^-\,\alpha}$ | $\mathcal{K}^{\#2}_{1^-\,\alpha}$ |
|---|---|---|---|---|
| $\mathcal{K}^{\#1}_{1^+}\dagger^{\alpha\beta}$ | 0 | 0 | 0 | 0 |
| $\mathcal{K}^{\#2}_{1^+}\dagger^{\alpha\beta}$ | 0 | 0 | 0 | 0 |
| $\mathcal{K}^{\#1}_{1^-}\dagger^{\alpha}$ | 0 | 0 | $-k^2\,\overset{(4)}{\kappa}_2$ | $-\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2$ |
| $\mathcal{K}^{\#2}_{1^-}\dagger^{\alpha}$ | 0 | 0 | $-\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2$ | $-2\,k^2\,\overset{(4)}{\kappa}_2$ |

| | $\mathcal{K}^{\#1}_{2^+\,\alpha\beta}$ | $\mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta}$ | $-9\,k^2\,\overset{(4)}{\kappa}_2$ | 0 |
| $\mathcal{K}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$ | 0 | 0 |

| | $\mathcal{J}^{\#1}_{0^+}$ | $\mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{J}^{\#1}_{0^+}\dagger$ | $\frac{1}{\mathrm{Det}(0^+)}$ | 0 |
| $\mathcal{J}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$ | 0 | $\frac{1}{3\,k^2\,\overset{(4)}{\kappa}_{15}}$ |

| | $\mathcal{J}^{\#1}_{1^+\,\alpha\beta}$ | $\mathcal{J}^{\#2}_{1^+\,\alpha\beta}$ | $\mathcal{J}^{\#1}_{1^-\,\alpha}$ | $\mathcal{J}^{\#2}_{1^-\,\alpha}$ |
|---|---|---|---|---|
| $\mathcal{J}^{\#1}_{1^+}\dagger^{\alpha\beta}$ | 0 | 0 | 0 | 0 |
| $\mathcal{J}^{\#2}_{1^+}\dagger^{\alpha\beta}$ | 0 | 0 | 0 | 0 |
| $\mathcal{J}^{\#1}_{1^-}\dagger^{\alpha}$ | 0 | 0 | $-\frac{1}{9\,k^2\,\overset{(4)}{\kappa}_2}$ | $-\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2}$ |
| $\mathcal{J}^{\#2}_{1^-}\dagger^{\alpha}$ | 0 | 0 | $-\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2}$ | $-\frac{2}{9\,k^2\,\overset{(4)}{\kappa}_2}$ |

| | $\mathcal{J}^{\#1}_{2^+\,\alpha\beta}$ | $\mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta}$ | $-\frac{1}{9\,k^2\,\overset{(4)}{\kappa}_2}$ | 0 |
| $\mathcal{J}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$ | 0 | 0 |

Abbreviations used in matrices

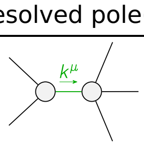
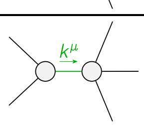
$$\Upsilon_1 == \overset{(4)}{\kappa}_1 + 7\,\overset{(4)}{\kappa}_2 \;\&\&\; \mathrm{Det}(0^+) == -\frac{3}{2}\,k^2\,(\overset{(4)}{\kappa}_1 + 7\,\overset{(4)}{\kappa}_2)$$

Lagrangian

$$\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + 2\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} +$$
$$9\,\overset{(4)}{\kappa}_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - 4\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 2\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + 9\,\overset{(4)}{\kappa}_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} -$$
$$6\,\overset{(4)}{\kappa}_2\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

| Added source term(s): | $\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$ |
|---|---|

| Source constraint(s) | # constraint(s) | Covariant form |
|---|---|---|
| $\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$ | 3 | $\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$ |
| $\mathcal{J}^{\#2\,\alpha\beta}_{1^+} == 0$ | 3 | $\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}$ |
| $\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == 0$ | 3 | $\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$ |
| $\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$ | 5 | $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ <br> $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} ==$ <br> $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ <br> $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$ |
| Total # constraint(s): | 14 | |

| Resolved pole(s) | # polarization(s) | Square mass | Residue |
|---|---|---|---|
|   | 1 | 0 | $\frac{3+2\sqrt{17}}{\overset{(4)}{\kappa}_2}$ |
|   | 1 | 0 | $\frac{3-2\sqrt{17}}{\overset{(4)}{\kappa}_2}$ |

| Resolved unitarity condition(s): | (Demonstrably impossible) |
|---|---|